Thomas R. Covert

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Biosketch

Thomas Covert is a Staff Data Scientist at Airbnb, working on pricing guidance modeling. He was previously an Assistant Professor at the University of Chicago Booth School of Business, the Scientific Director of the University of Chicago Energy and Environment Lab, and a Research Affiliate of the Energy Policy Institute at Chicago. He has sectoral expertise in the oil and gas industry, and extensive experience researching firm behavior in the industry. His research studies how oil and gas companies make investment decisions, and how regulatory policies and market institutions affect the quality of those decisions. He has leveraged state-level administrative data on oil & gas operations to study engineering improvements in hydraulic fracturing, tradeoffs between pipeline and rail transportation of crude oil, and the efficiency of the mineral lease market. Covert holds a PhD in Business Economics from Harvard University.

Education

Ph.D. Business Economics, Harvard University, 2008 - 2014

Dissertation: "Essays in Industrial Organization and Finance"

Committee: A. Pakes, B. Anand, P. Pathak, G. Lewis

S.B. Mathematics, Massachusetts Institute of Technology, 2001 - 2005.

Experience

Airbnb, Staff Data Scientist, May 2024 - present

University of Chicago Energy & Environment Lab, Scientific Director, July 2023 - May 2024

University of Chicago Booth School of Business, Assistant Professor of Microeconomics, July 2015 - June 2023

University of Chicago Department of Economics & EPIC, Postdoctoral Scholar, July 2014 - June 2015

Cornerstone Research (Boston), Senior Analyst, September 2005 - May 2008

Fields of Research Interest

Industrial Organization, Energy Economics, Finance, Applied Econometrics

Research

Peer-Reviewed Publications

1. Covert, T.R. and Ryan Kellogg (2026). Environmental Consequences of Hydrocarbon Infrastructure Policy. *Journal of Political Economy*
2. Covert, T.R. and Richard Sweeney (2023). Relinquishing Riches: Auctions vs informal negotiations in Texas oil and gas leasing. *American Economic Review*
3. Asquith, P., Au, A., Covert, T.R. and Pathak, P.A. (2013). The market for borrowing corporate bonds. *Journal of Financial Economics*

Working Papers

4. Covert, T.R., Hara, K., Kellogg, R. and Richard L. Sweeney (2026). Investment, Productivity, and Selection in the U.S. Shale Boom.
5. Covert, T.R. and Richard Sweeney (2022). Secrecy Rules and Exploratory Investment: Theory and Evidence from the Shale Boom.
6. Covert, T.R. and Richard Sweeney (2022). Winds of Change: Estimating Learning by Doing without Cost or Input Data.
7. Covert, T.R. (2015). Experiential and social learning in firms: the case of hydraulic fracturing in the Bakken Shale.
8. Asquith, P., Covert, T.R. and Pathak, P.A. (2019). The effects of mandatory transparency in financial market design: evidence from the corporate bond market.

Work in Progress

9. Covert, T.R., Kellogg, R.K. and Richard Sweeney (2022). Fracktime: A "Slack Time" theory of innovation and productivity growth in the U.S. shale business.
10. Covert, T.R., Kellogg, R.K. and Richard Sweeney (2022). Misallocation in U.S. Onshore Mineral Leasing.
11. Covert, T.R, Gazze, L., Greenstone, M. and Olga Rostapshova (2022). A Randomized Controlled Trial of Methane Emissions Reduction Policies.

Other Publications

1. Covert, T.R., Greenstone, M. and Knittel, C.R. (2016) Will We Ever Stop Using Fossil Fuels? *Journal of Economic Perspectives*

Conference and Seminar Presentations

Conferences: Conference on Empirical Industrial Organization at the University of Texas, Austin (April 2014), North American Summer Meetings of the Econometric Society (June 2014), CIRPEE Conference on Industrial Organization (July 2014), NBER Economics of Commodity Markets Meeting (May 2015), Oklahoma University Energy Finance Conference (September 2015), Midwest Energy Fest (May 2016, May 2017), AEA Annual Meeting (2017, 2018, 2019 and 2020), NBER Economics of Hydrocarbon Transportation Meeting (May 2017), AERE Summer Meeting (May 2017), Berkeley Energy Camp (2017, 2018 and 2019), NBER Summer Institute (July 2018 (IO)), CEBRA/Dallas Fed Workshop for Commodities (September, 2018), NBER Industrial Organization Winter Meeting (2020)

Invited Seminars: Cornell Economics (November 2014), Stanford GSB (November 2014), Princeton (February 2015), University of Wisconsin at Madison (April 2015), Federal Reserve Bank of Dallas (May 2015), Northwestern Economics (November 2015), University of Calgary Finance (December 2015), Wharton BEPP (April, 2016), Boston College (November, 2016), University of California Berkeley ARE (December, 2016), University of California Davis Economics (September, 2017), UT Austin (October, 2018), Wharton Energy Institute (November, 2018), Columbia SIPA (April, 2019), Microsoft Office of the Chief Economics (May, 2019), Northwestern Economics (May, 2019), Federal Reserve Bank of Kansas City (July, 2019), Cornell Economics (February, 2020)

Professional Activities

Reviewer for *American Economic Review*, *AEJ: Economic Policy*, *AEJ: Microeconomics*, *AEJ: Applied Economics*, *Journal of Political Economy*, *International Journal of Industrial Organization*, *Agricultural*

and Resource Economics Review, *Journal of Environmental Economics and Management*, *Econometrica*, *Energy Journal*, *Quarterly Journal of Economics*, *Review of Economics and Statistics*, *Journal of Public Economics*, *Journal of Industrial Economics*, *Review of Economic Studies*

Honors, Scholarships and Fellowships

Colorado Oil and Gas Conservation Commission, "Investigating Novel Techniques to Reduce Methane Emissions" 2020 (co-PI, $2,000,000)

NSF Grant #1919499, "Productivity, Efficiency, and Distortions in the U.S. Shale Boom", 2019 (PI, $536,738)

Joel F. Gemunder Fundation Grant to Study Productivity in the U.S. Shale Business, 2018 (co-PI, $275,000)

Harvard Environmental Economics Program *Enel Endowment Prize* (2014)

Sandra Ohrn Family Graduate Dissertation Fellowship, Harvard University (2013)

Doctoral fellowship, Harvard Business School (2008-2013)

Last updated: October 25, 2026